

Title of Report

A National Airspace System Reference Architecture as a System of Systems

ISD Capstone Project Report

Submitted to:

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Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 5.0 Impact/Financial Benefits), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report.

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1 Introduction and Current State

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

My project aims to build a reference MBSE architecture for how airspace intent is managed across the NAS — spanning the organizations that define and enforce rules, the domains managing airspace resources, the systems supporting flight operations, and the onboard systems that turn intent into executable trajectory and guidance behavior.

In his 2005 paper, DeLaurentis states the following: [delaurentis-2005-understanding-transportation-as-a-system-of-systems-design-problem.pdf](#) "PROBLEMS of significant complexity are facing decision-makers within government and industry, and the moniker of "system-of-systems" (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the "stovepipe" mode- their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems, exemplified in this paper by future national transportation system realizations. A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time. 1 And this need is urgent, for the SoS type almost always involves decisions that commit large amounts of money, for which ultimate failure or success carries heavy consequences over several generations, and that impact large segments of the public. The costs associated with a chosen path may still be high, but the intent is to maximize the probability that these costs result in good consequences rather than bad."

While this reference is from 2005, it is still relevant today. The NAS is a complex system of systems, and the decisions made by the various stakeholders have significant consequences for the safety, efficiency, and sustainability of the air transportation system.

According to deCasto, Airline companies spend time and money in creating optimal operational plans to maximize the revenue. However, at the day of operation, the unpredictable events (bad weather, aircraft malfunctions and crew absenteeism, for example) might change completely that operational plan and, consequently, the revenue objectives behind it (Clausen et al., 2010). To deal with this problem, airline companies have a special department called Airline Operations Control Center (AOCC) that includes teams of human experts specialized in solving problems related to aircraft, crew members and passengers, in a process called disruption management. The AOCC is responsible for monitoring the airline's operations and making real-time decisions to minimize the impact of disruptions on the airline's schedule and revenue objectives. This includes re-routing flights, rescheduling crews, and communicating with passengers about delays or cancellations.

The costs associated with these disruptions can be significant, both in terms of financial losses and customer satisfaction (cite specific sources; several include data). Initial flight plannings include selections of altitudes, routes, and speeds that are optimized for fuel efficiency, time, and cost. However, there is no standard formulation of "cost" in the literature, and it can include a variety of factors such as fuel consumption, crew costs, maintenance costs, passenger compensation, and lost revenue from missed connections or cancellations. When disruptions occur, the AOCC must make trade-offs between these factors to minimize the overall cost to the airline. When disruptions occur, these plans may need to be adjusted, leading to increased fuel consumption, longer flight times, and additional costs for the airline. Additionally, passengers may experience delays or cancellations,

leading to dissatisfaction and potential loss of future business.

In this paper, I will explore the value tradeoffs that aviation stakeholders experience when disruptions occur, and solutions are employed with various levels of myopia. When a dispatcher or AOCC team member makes a decision, they may be focused on the immediate impact of that decision (e.g., minimizing delay for a specific flight) without considering the broader implications for the airline’s overall operations or the NAS as a whole. Moreover, cost models employed today typically focus on easily quantified factors, including fuel consumption and crew costs, while other factors, such as maintenance cost, passenger satisfaction or environmental impact, may be more difficult to quantify and are often overlooked. This can lead to suboptimal outcomes, such as increased congestion in certain airspace sectors, missed opportunities for more efficient routing, or decreased overall profits driven by customer dissatisfaction. By modeling the NAS as a system of systems and analyzing the propagation of operational intent, I aim to identify strategies for improving decision-making and reducing the negative impacts of disruptions on the air transportation system.

1.1 Overview of the Project

Plan ref: to-do-list.md S1 — Describe the industry/environment (commercial aviation and the NAS), the social context, and the key stakeholders (preview – full analysis in Section 1.3). State the project’s research questions from Project_To-Do List.md S1 (“Finalize project research questions”) and explain why they matter. Draw on README.md’s framing of the trajectory-intent lifecycle (Strategic objective -> mission plan -> flight plan -> ATC constraints -> trajectory negotiation -> FMS intent -> guidance commands -> aircraft motion) as the throughline.

1.2 Review of Current State Design / Literature Review

Plan ref: to-do-list.md S2–S4 — Synthesize (do not merely list) findings from the three literature-review tracks: airline planning and operations (schedule planning, fleet assignment, crew scheduling, maintenance routing), day-of- operations and turnaround (A-CDM, weight and balance), and OCC/dispatch and flight execution (disruption management, dispatcher responsibilities, nominal ATC/IFR flow). Pull from per-source annotations captured via the literature-review workflow into **evidence/source-register.md** rather than re-deriving them here. Identify the research gap this capstone addresses (see also Project_To-Do List.md S16 “Research gap”).

NOTE: this subsection is a compressed synthesis of the underlying literature annotations, not a restatement of every “identify X” checklist item in Project_To-Do List.md S2–S4 – those live in the source register and feed this section, not the other way around.

1.3 Stakeholders

Plan ref: to-do-list.md S7 — Identify key project stakeholders (individuals/organizations affected by or perceiving themselves affected by NAS operational decisions). Draw on the PESTLE stakeholder discovery, stakeholder register, and RACCI matrix from Project_To-Do List.md S7. Confirm the stakeholder set is diverse (airline, passenger, ATC/ANSP, airport, flight crew, environmental/ societal, military – see S8’s per-stakeholder objective categories) and that stakeholders with little formal power (e.g., passengers, communities affected by noise/emissions) are represented.

1.4 Scope

Plan ref: to-do-list.md S1 — State what is explicitly included and excluded from the System of Interest (SOI), the level(s) of abstraction modeled, and the criteria used to draw system and stakeholder/actor boundaries. This section should resolve the open boundary questions currently tracked in `knowledge/questions/open-questions.md S1` – if any remain open when this section is drafted, that is a signal the SOI work in `Project_To-Do List.md S1` is not yet closed.

1.5 Project Deliverables

Plan ref: to-do-list.md S10, S14, S16 — State the goals and deliverables: the SysML/Cameo reference architecture, the traceability chain from stakeholder needs to aircraft-level behavior, and the optimization-demonstration capability (Section 3). Include the quantifiable measures used to assess success (e.g., traceability coverage, stakeholder coverage, demonstration of a local-vs-SoS optimization tradeoff).

1.6 Project Timing

Plan ref: to-do-list.md S1 (new) — Revisit the timeline from the 503 Proposal submittal (see `prework/ISD 503 Submittal.pdf`) and list the sequence of tasks – keyed to `Project_To-Do List.md`’s 16 sections – with target completion dates. A “Project Timing” task did not previously exist in the plan; it has been added to `Project_To-Do List.md S1` this session so this subsection has something concrete to draw from.

2 Assumptions and Methodology

Plan ref: to-do-list.md S1 — State the working assumptions carried through the project – see `README.md` “Working Assumptions” and `projects/nas-sos-capstone/index.md` “Working assumptions (guardrails)” (e.g., emphasis on structure/behavior/interfaces/traceability rather than optimization math; decision-support/optimization treated as a capability inside the architecture, not the whole subject).

Plan ref: to-do-list.md S6 — Describe the methodology for exploring and comparing alternative system decompositions (organization/authority-based, lifecycle/process-based, physical-system, information-flow, decision-authority, intent-to-trajectory) and the criteria used to evaluate them (what each exposes/obscures, where boundaries are ambiguous, which best supports optimization-impact analysis).

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in Cameo: package organization, diagram types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs -> objectives -> requirements -> systems -> activities -> information exchanges -> aircraft behavior). Note per `workflows/update-architecture.md` that Cameo is the source of truth and the XML export is a generated artifact.

Plan ref: to-do-list.md S11 — Describe the optimization-study methodology: scenario selection criteria, system-state and decision-variable definitions, constraint definitions, the multi-objective formulation $J = w_1 J_1 + w_2 J_2 + \dots + w_n J_n$, normalization approach for objectives with different units/scales, and the stakeholder weighting scheme. This section should be detailed enough that the scenario and formulation could be reproduced.

3 Results and Discussion

Plan ref: to-do-list.md S5 — Present the nominal-flight ConOps (strategic/commercial planning through day-of-operations, turnaround, flight execution, and postflight/continuation) as the operational backbone the architecture models. Summarize; put the full activity breakdown in Appendix A.1.

Plan ref: to-do-list.md S8 — Present the stakeholder objective/cost/value ontology: what "optimal" means per stakeholder (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military), and the classification of each objective as hard constraint, optimization objective, cost/penalty, MOE, or MOP. Highlight conflicting vs. aligned objectives and objectives with externalized costs.

Plan ref: to-do-list.md S9 — Present the myopic-optimization analysis: the definition of "local optimum" in the SoS context, candidate SoS measures of value, and the documented conflicts (decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence). See the flagged scope note below.

Scope note (flag, not a TODO to resolve silently): Project_To-Do List.md S9 catalogs eight candidate local-vs-SoS conflicts, each with eight documented attributes (64 cells). A 15–40 page report body cannot carry all eight in full; plan to present 2–4 illustrative conflicts in the main body (favoring whichever one becomes the S11–S14 demonstration scenario) and move the remaining catalog entries to Appendix A.4.

Plan ref: to-do-list.md S10 — Present the SysML architecture: context diagram, package structure, stakeholder model, BDDs, IBDs, information-object model, interface/item-flow model, nominal-flight activity diagram, responsibility swimlanes, and requirements model with traceability. Embed the most illustrative diagrams here; move the full diagram set to Appendix A.3.

Plan ref: to-do-list.md S12, S13 — Present the optimization/simulation results: the base-line scenario, single-stakeholder vs. balanced/SoS optimization outcomes, and how simulation entities/decisions/objectives/constraints trace back to SysML blocks, properties, goals, and requirements. Show how the architecture exposes impacts a standalone optimizer would omit (S13's key point).

Scope note: Project_To-Do List.md S12 as written (Pareto fronts, full objective-weight sweeps, sensitivity analysis, tipping-point identification across the whole experiment matrix) reads as a full optimization research study layered on top of the architecture work. Per this project's own working assumption ("not an optimization paper"; optimization is a capability inside the architecture, not the whole subject) and the 15–40 page report constraint, consider scoping S12's demonstration to one representative scenario with a single weight sweep and one Pareto-style comparison, rather than the full matrix – see the note added to `knowledge/questions/open-questions.md` under "Optimization study scope (S11)."

4 Recommendations

Plan ref: to-do-list.md S14, S15 (S16 bullet added this session) — Restate the main findings from Section 3 and provide recommendations for moving forward. Include the alternatives considered (e.g., alternative decompositions from S6, alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

NOTE: Project_To-Do List.md did not previously have a distinct "Recommendations" deliverable separate from "Conclusions and future work" – added to S16 this session so this section has an explicit home in the plan.

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its optimization-demonstration capability. Translate the S9 conflict analysis and S12 demonstration results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, gate/runway utilization) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to S8’s “objectives whose costs are externalized onto another stakeholder” and S9’s conflict catalog).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template’s sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a “before” baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Project_To-Do List.md did not previously have a task that produces a quantified/monetized impact figure – an explicit “Impact / quantified benefits” bullet was added to S16 this session so at least one deliverable targets this section directly. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

6 Summary/Conclusions

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings/ recommendations, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., “reduces network-level delay propagation by an illustrative X% under the demonstrated weighting” rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the optimization demonstration (S12 scope note, Section 3), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

8.0 References

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A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in Section 3: full BDD/IBD set, requirements model, and complete traceability matrices (needs-¿objectives-¿requirements-¿systems-¿activities-¿information exchanges-¿aircraft behavior).

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 3.

A.6 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (evidence/source-register.md): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.